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Analyzing the interrelationships of evaluation indicators in the open data services industry's efforts toward digital transformation: A novel group decision-making approach

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ABSTRACT

The open data services industry is critical in advancing artificial intelligence (AI), particularly as businesses navigate digital transformation. This transformation is essential for enhancing AI-driven decision-making, fostering innovation, and improving operational efficiency. This study aims to identify and analyze the interrelationships among key evaluation indicators to optimize performance within the open data services industry. Using data collected from expert interviews and a combined Delphi method with the interval single-valued trapezoidal neutrosophic DEMATEL (DIN-DEMATEL) technique, we establish influence relationships and prioritize the weights of these indicators. Our analytical approach provides a structured evaluation of how criteria interact and influence one another. The findings highlight "resource utilization efficiency," "innovation-driven productivity," and "financial performance and sustainability" as the three most influential indicators in the overall evaluation system. These indicators drive efficiency, foster innovation, and ensure financial stability, forming the foundation for operational excellence and strategic growth. The findings provide a clear framework for businesses aiming to enhance their competitiveness through digital transformation while advancing the theoretical understanding of interrelationships within the open data services industry. The study offers actionable recommendations for companies to target improvements in these areas, enhancing competitiveness and resilience in digital transformation.

1. Introduction

Open data services, an increasingly vital concept for governments and enterprises worldwide, make datasets publicly accessible online in open formats, allowing anyone to freely use, reuse, and share these data (Mohamad et al., 2024). This approach enhances transparency and provides a critical resource for the research and development of artificial intelligence (AI) technologies. As data volume grows exponentially, open data services have emerged as a key enabler for AI advancements (Pencheva).

The transformative impact of open data services on AI development is evident in several key areas: they foster the creation and training of AI models, accelerate AI innovation and application, and enhance transparency and accountability in AI operations (Madan & Ashok, 2023). Firstly, open data services supply a vast and diverse dataset pool essential for training accurate and efficient AI models, particularly in machine learning and deep learning. These large-scale datasets, such as those made publicly available by government agencies—spanning transportation, health, and environmental data—play a crucial role in developing predictive models and decision support systems (Kusumastuti).

Open data services accelerate AI innovation by providing publicly accessible and reusable data, fostering interdisciplinary collaboration and enabling cost-effective AI development (Neumann et al., 2024; Van Noordt & Misuraca, 2022). These services stimulate entrepreneurship and innovation by lowering data acquisition barriers and allowing startups and researchers to explore new AI applications. Beyond innovation, open data services enhance AI transparency and accountability.

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Transparency is crucial for public trust, as making AI datasets and processes publicly accessible improves understanding and acceptance of AI decision-making (Madan & Ashok, 2023). Additionally, transparency facilitates the identification of biases and fairness issues, promoting responsible AI use (Basole et al., 2024). Open data services support AI research, accelerate technological advancements, and ensure ethical AI deployment. To fully unlock their potential, governments and businesses must continue advocating for data openness, ensuring AI development remains transparent, inclusive, and innovation-driven (Holmström).

With the rapid advancement of digital technologies, digital transformation has become a core strategy for businesses to ensure continuous growth and maintain competitiveness (Gobble, 2018). For the open data services industry, embracing digital transformation enhances service efficiency and quality, opens up new business opportunities, and generates additional value (Wu et al., 2021). Open data service providers can significantly improve their data processing and analysis capabilities by adopting cutting-edge technologies such as cloud computing, big data analytics, AI, and machine learning. Cloud technology enables flexible and scalable data storage and management while ensuring data security and privacy (Holmström). Moreover, combining big data and AI allows providers to extract valuable insights from large datasets, offering more personalized and precise services (KitsiosKamariotou). Furthermore, digital transformation facilitates collaboration and integration among open data service providers, other industries, and organizations by developing open APIs and digital platforms (Lee et al., 2023). This expands the scope and impact of services and provides users with more diverse and convenient data services. By connecting with partners from various fields, providers can explore new business models and revenue streams (Rana & Daultani, 2023).

The benefits of digital transformation for open data service providers are evident (Wen et al., 2023). First, digital operations greatly enhance business efficiency and service quality while reducing the consumption of human and material resources. Second, using digital technologies allows providers to offer more diverse and value-added services, increasing customer satisfaction and loyalty (Struijk et al., 2023). Additionally, digital transformation broadens market reach and expands the customer base, boosting competitiveness. Finally, digital data collection and analysis enable providers to gain deeper insights into market trends and customer needs, leading to more accurate and effective business decisions (Holmström; Rana & Daultani, 2023). Thus, digital transformation is not merely a trend but an essential strategic choice for open data service providers. Through digitalization, providers can enhance operational efficiency and secure a competitive foothold in the market, creating new business value (Zoppelletto et al., 2023).

While previous studies provide guidance on the digital transformation of the open data services industry, several critical gaps remain.

(i) Most research examines only a limited number of indicators, lacking a systematic analysis of their interrelationships across multiple dimensions. Effective methods for identifying and quantifying these interdependencies remain underexplored.
(ii) Few studies address the ambiguity, uncertainty, and consensus issues inherent in expert evaluations, leading to gaps in reliability and robustness.
(iii) While prior research has identified key factors influencing digital transformation, it often lacks prioritization and weight analysis of these indicators, limiting the practical applicability of findings in real-world operations.

This study aims to bridge these gaps by developing a comprehensive evaluation framework tailored to the open data services industry. Unlike prior research, which focuses on isolated factors, this study employs a systematic approach to analyze the interdependencies among key evaluation indicators and their impact on industry competitiveness. Using a novel group decision-making model, this research offers actionable insights for industry stakeholders. It provides a robust framework for optimizing strategies in service innovation, operational efficiency, and long-term sustainability while supporting AI-driven advancements. This dual focus on theoretical contributions and practical applications ensures relevance for both academia and industry, guiding policymakers, business leaders, and researchers in driving digital transformation.

This study adopts a structured multi-phase methodology to ensure a comprehensive and reliable analysis. First, a literature review and expert interviews identify key evaluation indicators, categorized into five dimensions: data governance and transparency, operational efficiency, system resilience, innovation, and collaboration. Next, multiple experts assess interrelationships among these dimensions and indicators through a six-round influence determination process, starting at the dimension level and refining at the indicator level. Finally, the study employs the Delphi and the interval single-valued trapezoidal neutrosophic DEMATEL (DIN-DEMATEL) technique to quantify influence relationships and weights. This methodological approach is chosen for its ability to handle uncertainties and ambiguities in expert evaluations, ensuring robust, consensus-driven results. DIN-DEMATEL enhances analytical precision by integrating fuzzy logic and neutrosophic sets, making it particularly suited for complex decision-making. The multi-phase structure provides a systematic and actionable framework for the open data services industry, helping stakeholders prioritize resources, strengthen decision-making, and drive sustainable digital transformation.

This study makes groundbreaking contributions to advancing the open data services industry, particularly in the context of digital transformation and the challenges presented by the era of big data. This research offers a novel approach to understanding the intricate factors driving industry growth and competitiveness by developing a comprehensive evaluation framework and leveraging cutting-edge group decision-making methodologies. It provides actionable insights that empower stakeholders to navigate digital transformation effectively while enhancing the industry's resilience and adaptability. This study makes four key contributions.

(i) It constructs a detailed evaluation framework that integrates critical dimensions of digital transformation in the open data services industry.
(ii) It enhances the reliability of expert assessments by applying the Delphi method, eliminating extreme opinions and ensuring balanced expert consensus.
(iii) It moves beyond identifying key factors by uncovering interdependencies and relative importance among indicators, offering a data-driven perspective that enriches both theoretical understanding and practical applications in the big data era.
(iv) It delivers strategic recommendations tailored to the industry's digital transformation challenges, equipping stakeholders with the tools to prioritize key indicators, foster innovation, and enhance competitiveness in a rapidly evolving digital landscape.

2. Literature review

This section provides a comprehensive review of the open data services industry, focusing on its current developments and challenges within the context of digital transformation. Additionally, the section introduces the proposed evaluation framework, which is designed to assess the industry's performance by integrating multiple dimensions and indicators.

2.1. Open data service

In recent years, the continuous growth of data has prompted companies to seek more effective methods for extracting valuable insights to enhance competitiveness (Kitsios and Kamariotou, 2023). This demand

has given rise to the data services sector, which focuses on maximizing the value of data through a range of products, services, and solutions (Matheus et al., 2020). In the digital era, businesses increasingly adopt data-driven decision-making by leveraging data analysis to inform strategies (Gong & Li, 2023). However, many companies struggle to fully utilize collected data due to a lack of skilled personnel in data processing and analysis. Paparova et al. (2023) highlight that in today's rapidly evolving IT landscape, companies are integrating external knowledge to drive innovation and reduce costs. Open data has emerged as a key resource for predicting market trends and meeting customer needs, enabling businesses to offer innovative products and services that create competitive advantages (Shao, 2024). Governments also support this by providing open data through various platforms, facilitating its integration for business intelligence, process optimization, and R&D. This not only helps in developing innovative business models but also enhances value for both businesses and customers (Schneider et al., 2023).

In a data-driven business environment, effective management and application of open data are essential for enhancing competitiveness and fostering innovation. Establishing robust data governance strategies is crucial to ensuring data quality and security while unlocking the latent value of data to support decision-making (Jarvenpaa and Essén, 2023). Additionally, collaboration with public and private sectors in acquiring, sharing, and applying open data expands its applications and strengthens a company's agility in responding to market changes (Gao et al., 2023). Digital transformation is an ongoing process that requires tailored strategies based on a company's stage in the transformation journey (Zoppelletto et al., 2023). In the initial stages, the focus should be on building infrastructure, such as information systems and data management capabilities, to reduce information asymmetry and enhance decision-making efficiency. Typically, businesses progress through three phases: digitization, optimization, and transformation. Rana and Daultani (2023) emphasize that in manufacturing, digital transformation involves leveraging big data for predictive models and integrating upstream and downstream data. This approach enables precise production and inventory control, reducing costs and risks while boosting operational efficiency.

In the digitization phase, industries transition from paper-based and manual processes to digital systems, enabling efficient data storage and management (Lee et al., 2023). In the digital optimization phase, companies leverage digital tools and information to enhance product quality, management efficiency, and internal processes. Digital transformation, the final phase, involves integrating digital technologies across all business functions, fundamentally reshaping operations and business models (Kraus et al., 2023). Technologies such as robotic process automation, AI, machine learning, and blockchain play pivotal roles in transforming traditional supply chain management and boosting efficiency (HHolmström, 2022). Additionally, big data, cloud computing, and the Internet of Things (IoT) enhance supply chain visibility and promote process standardization and automation, accelerating digital transformation (Wu et al., 2021). A broad perspective is essential for successful digital transformation, requiring the integration of corporate strategy, organizational structure, and digital technologies. This approach focuses on redefining products and services based on customer needs and promoting cross-departmental collaboration to drive innovation (Pinto et al., 2023). As companies advance, adopting data-centric business models that integrate open and internal data is crucial for identifying new opportunities (Struijk et al., 2023). Strong data processing capabilities, along with AI, machine learning, and big data, are vital for enhancing efficiency and accuracy (Janssen et al., 2020). Engaging in open data ecosystems and forming cross-industry partnerships further fosters innovation and product diversity (Madan & Ashok, 2023). Ultimately, the open data services industry presents new business models and opportunities but demands stronger data governance, innovation, and collaboration. To succeed, companies must continuously explore and harness open data to drive sustainable growth and competitiveness.

2.2. The proposed evaluation framework for open data service industrial

This study conducted an extensive literature review and in-depth expert interviews to identify five core dimensions that capture the key challenges and opportunities of the open data services industry in digital transformation. The selected experts, detailed in Section 4, are all representative figures in the open data services industry, ensuring the relevance and credibility of their input. The core dimensions include Data Governance and Transparency, Operational Efficiency and Productivity Enhancement, System Resilience and Risk Management, Innovation and Market Expansion, and Collaboration and Co-Creation of Value. Based on these dimensions, this study further screened specific indicators through empirical studies and theoretical models found in the literature, focusing on those that have been widely discussed and clearly defined. This ensured that the selected indicators could be effectively applied in the context of open data services industry.

To ensure the applicability of the indicators, this study conducted multiple rounds of expert interviews with professionals from the open data services industry, data governance field, and digital transformation sector. These experts provided feedback on the initially selected indicators, which helped refine and validate the accuracy and relevance of each indicator. The final set of indicators was informed by theoretical foundations and practical insights, ensuring that they accurately reflect the key factors influencing the industry's development. In the end, this study established 24 specific indicators (Table 1), each with a clear definition and supported by relevant references from the literature, providing a solid foundation for the subsequent research.

2.2.1. Data governance and transparency

Data governance is essential for managing how organizations acquire, utilize, and make decisions based on data (Gegenhuber et al., 2023). Collaboration between the public and private sectors enhances data applications and quality through cleaning, refinement, standardization, and bias prevention (Slimani et al., 2024). Additionally, government policies on data standards and licenses guide organizations in implementing effective governance practices. Improving data interoperability is crucial for seamless data integration and exchange across systems, requiring standardized metadata management (Wirtzüller, 2023). Another key factor is data circulation, where continuous feedback between data suppliers and users enhances quality and usability. Effective data governance must be industry-specific, ensuring data is formatted and structured to meet diverse application needs (Karkošková, 2023).

In disaster risk management, Lee and Roh (2023) also emphasize that fostering added-value applications of open data and improving its usability requires strong collaboration and interaction between the public and private sectors. In digital transformation, organizations must recognize data as a valuable asset, clearly defining data access rights, application models, and monitoring rules to maximize data value while minimizing costs and risks. Chen et al. (2023) identify eight key evaluation factors of data governance within the open data ecosystem, with data quality, government policies, data interoperability, data circulation, and demand matching being particularly important. Buhalis et al. (2023) further elaborate on critical activities for enhancing data management maturity from an ecosystem perspective, stressing the need to establish data standards based on the needs of data consumers and users. Infrastructure and tools can be developed through stakeholder collaboration to maximize business efficiency.

2.2.2. Operational Efficiency and Productivity Enhancement

Operational efficiency and productivity enhancement in a business ecosystem are defined as transforming innovation into cost advantages, new products, and enhanced functionalities (Thomas & Ritala, 2022). Within such ecosystems, collaboration, competition, and

Table 1
The proposed dimensions and indicators for constructing an AI business ecosystem in the open data services industry.

Dimension	Indicator	Definiation	Reference
Data Governance and Transparency (A)	Data Quality (A1)	This indicator refers to the accuracy, completeness, consistency, and reliability of data. High-quality data is essential for effective governance and transparency, as it ensures that stakeholders can trust and use the data for decision-making without concerns about its integrity or validity.	Gegenhuber et al. (2023); Perdana et al. (2025)
	Government Policies (A2)	This indicator encompasses the regulations, standards, and legal frameworks set by government entities to ensure data security, privacy, and transparency. These policies play a critical role in guiding organizations to adopt effective data governance practices and promote the responsible use of open data.	Janssen et al. (2020); Karkošková (2023)
	Data Interoperability (A3)	This indicator is the ability of different systems and organizations to exchange, integrate, and use data efficiently across various platforms. It ensures seamless data sharing, which enhances transparency and allows for broader, more effective use of data in cross-organizational and cross-sectoral contexts.	Jarvenpaa and Essén (2023); Gegenhuber et al. (2023)
	Data Circulation (A4)	This indicator refers to the continuous flow of data between providers and users, enabling iterative feedback to improve data quality and applicability. This process facilitates dynamic data governance by ensuring that data is regularly updated, refined, and adjusted to meet evolving needs.	Janssen et al. (2020); Karkošková (2023)
	Demand Alignment (A5)	This indicator measures the extent to which data meets the specific needs and requirements of its users. Effective alignment ensures that data governance processes are oriented toward delivering value to stakeholders by providing data that is relevant, timely, and fit for purpose.	Gegenhuber et al. (2023); Perdana et al. (2025)
Operational Efficiency and Productivity Enhancement (B)	Resource Utilization Efficiency (B1)	This indicator refers to the ability to maximize the use of available resources, such as labor, capital, and technology, in producing goods or services. Efficient resource utilization leads to higher productivity by minimizing waste and optimizing operational processes, directly enhancing the overall operational efficiency.	Gong and Li (2023); Xiang et al. (2024)
	Value Creation Productivity (B2)	This indicator measures the ability of organizations to generate value through the effective coordination and integration of resources, skills, and knowledge. It reflects how well a company can translate its inputs into value-added products or services that enhance competitiveness in the market.	Babu et al. (2020); Bitencourt et al. (2024)
	Innovation-Driven Productivity (B3)	This indicator focuses on the capacity to improve operational processes and develop new products or services through the implementation of cutting-edge technologies and creative solutions. This indicator emphasizes how innovation contributes to enhancing both efficiency and competitiveness.	Zimmerling and Chen (2021); Park and Gil-Garcia (2022)
	Financial Performance and Sustainability (B4)	This indicator evaluates the financial health and profitability of an organization, emphasizing the long-term sustainability of its operations. It includes the ability to manage costs, generate revenue, and maintain financial stability, all of which are essential	Bohnsack et al. (2024); Karkošková (2023)

(continued on next page)

Table 1 (*continued*)

Dimension	Indicator	Definiation	Reference
		for supporting sustained operational efficiency and productivity growth.	
System Resilience and Risk Management (C)	Predictive Capability (C1)	This indicator refers to the ability of a system to forecast potential disruptions, market changes, and risks. Strong predictive capability allows organizations to proactively identify and mitigate risks, ensuring that they can prepare for future uncertainties with effective strategies.	Carayannis et al. (2025); Tiwari et al. (2024)
	Network Stability and Structural Robustness (C2)	This indicator assesses the resilience of an organization's network structure, emphasizing the strength and stability of its relationships with partners, suppliers, and other stakeholders. A stable and robust network ensures continuity and minimizes the impact of disruptions by providing reliable support and resource-sharing mechanisms.	Lin and Lo (2024); Tiwari et al. (2024)
	Adaptive Capacity (C3)	This indicator measures the ability of an organization to quickly adjust its operations, processes, and strategies in response to changing external conditions or unforeseen challenges. It is critical for maintaining operational efficiency in the face of disruptions and shifts in the business environment.	Lin and Lo (2024); Zhang (2025)
	Recovery Capability (C4)	This indicator focuses on the organization's ability to recover from disruptions or crises, restoring normal operations swiftly and effectively. Strong recovery capability ensures minimal downtime and loss, enabling the organization to maintain continuity and competitiveness in the face of challenges.	Carayannis et al. (2025); Tiwari et al. (2024)
	Strategic Flexibility and Timely Adjustment (C5)	This indicator refers to the ability of an organization to make timely and strategic decisions regarding the adjustment or elimination of outdated processes, technologies, or partnerships. Proactively removing or replacing underperforming elements helps maintain resilience and keeps the organization agile and responsive to new opportunities.	
Innovation and Market Expansion (D)	Visionary Leadership (D1)	This indicator refers to the ability of leaders to inspire innovation and drive strategic direction within an organization. Leaders who can effectively identify market trends and harness new technologies play a crucial role in fostering innovation and leading the organization towards new market opportunities.	Park and Gil-Garcia (2022); Afeltra et al. (2023)
	Proactive Engagement (D2)	This indicator reflects the active participation of organizational members and stakeholders in innovation processes. High levels of engagement are essential for driving collaborative innovation, ensuring that ideas are continuously generated and implemented to create competitive advantages in the market.	Kitsios and Kamariotou (2023); Bitencourt et al. (2024)
	Product and Service Differentiation (D3)	This indicator measures an organization's ability to develop unique and innovative products and services that stand out in the market. Differentiation through innovation is key to capturing new market segments and maintaining a competitive edge in a rapidly evolving environment.	Zimmerling and Chen (2021); Silva et al. (2024)
	Leverage of Intangible Assets (D4)	Intangible assets such as intellectual property, brand reputation, and knowledge capital are critical resources that drive innovation and market	Afeltra et al. (2023); Kitsios and Kamariotou (2023)

(*continued on next page*)

Table 1 (*continued*)

Dimension	Indicator	Definiation	Reference
		expansion. The ability to effectively leverage these assets enhances an organization's capacity to innovate and enter new markets.	
	Integration with External Networks (D5)	This indicator emphasizes the importance of building and maintaining strong relationships with third-party networks, including partners, suppliers, and industry collaborators. Such networks provide access to new resources, knowledge, and technologies, which are essential for driving innovation and expanding market reach.	Gegenhuber et al. (2023); Silva et al. (2024)
Collaboration and Co-Creation of Value (E)	Responsiveness to Market Demands (E1)	This indicator measures an organization's ability to swiftly and effectively respond to changes in market demands through collaborative efforts. It reflects the agility of businesses and their partners in adapting to customer needs and industry shifts, thus ensuring continued relevance and value creation.	Babu et al. (2020); Tiwari et al. (2024)
	Resource Complementarity (E2)	This indicator refers to how well the resources of different partners within an ecosystem align and complement one another. Effective collaboration involves leveraging the strengths and unique capabilities of each partner to co-create value that could not be achieved individually.	Gong and Li (2023); Tiwari et al. (2024)
	Knowledge and Experience Sharing (E3)	The exchange of knowledge, expertise, and experiences between partners is crucial for fostering innovation and driving value creation. This indicator highlights the importance of collaborative learning and knowledge transfer in enhancing the overall capabilities of the ecosystem.	Babu et al. (2020); Bitencourt et al. (2024)
	Resource Integration Capability (E4)	This indicator is the ability to effectively combine and coordinate various resources from different partners to achieve common goals. Strong integration capability enables organizations to optimize resource use and enhance the value created through collaboration.	Gong and Li (2023); Tiwari et al. (2024)
	Collaborative Value Proposition (E5)	This indicator assesses the strength of the value proposition developed through collaborative efforts. It reflects how well partners co-create a compelling value offering that differentiates them in the market and delivers significant benefits to customers and stakeholders.	Gong and Li (2023); Bitencourt et al. (2024)

complementary interactions among members drive innovation and optimize resource utilization, enhancing overall competitiveness and productivity. Productivity in this context involves several key aspects, such as the efficiency of converting resources like capital and labor into products and services, the coordination and deployment of resources through collaboration to create and capture more excellent value, and the capacity to leverage new technologies to develop innovative solutions. Maintaining a healthy financial standing and profitability ensures the ecosystem's sustainable operation (Bohnsack; Robertson et al., 2020). These elements collectively illustrate how ecosystem members interact to improve operational efficiency and effectiveness. In the digital transformation era, such interactions are further enhanced by adopting advanced technologies, such as AI, big data, and automation, which streamline operations, reduce costs, and boost productivity. Digitalization enables organizations to respond more rapidly to market demands, improving service quality and customer satisfaction (Silva et al., 2024).

2.2.3. *System resilience and risk management*

In today's dynamic industrial landscape, businesses must navigate challenges in R&D, operations, risk management, and partnerships (Xue et al., 2024). Success depends on building a resilient business ecosystem through strategic planning, resource allocation, and adaptability Key resilience factors include member survival capability, ecosystem sustainability, trend predictability, elimination mechanisms, and continuity. Knowledge transfer, innovation, and network stability are crucial for maintaining ecosystem vitality (Senyo et al., 2019). A healthy business ecosystem requires sustainability, adaptability, innovation, and renewal capacity, often supported by crowdsourcing for risk mitigation (Suuronen et al., 2024).

Resilience extends beyond structure to market responsiveness and recovery capability (Basole et al., 2024). Predicting risks, establishing stable yet flexible networks, and swiftly resuming operations after disruptions are essential. Additionally, the timely elimination of underperforming members and integration of adaptive ones ensures long-term ecosystem health (Fernández-Portillo et al., 2024). Businesses must

continuously strengthen resilience strategies to sustain competitive advantage and innovation in an ever-evolving market.

2.2.4. Innovation and Market Expansion

Innovation and market expansion are critical dimensions for evaluating whether businesses and their partners within a business ecosystem possess the capability to leverage emerging technologies and knowledge to develop new products and services, thereby creating a competitive advantage in the market. This dimension encompasses key indicators such as organizational diversity, as well as the diversity of products and technologies (Dzhengiz and Patala, 2024).

In a business ecosystem, leaders must demonstrate visionary leadership, actively participating in ecosystem activities and engaging with other members to foster innovative thinking (Lee & Sohn, 2019). The driving force behind innovation lies in integrating, exchanging, and applying resources, combining the knowledge, skills, experiences, and services of various members (Whittaker et al., 2023). Furthermore, the diversity of products and services within an ecosystem is built upon the foundation of open sharing and knowledge exchange among stakeholders. The synergy within the ecosystem enhances the ability of stakeholders to access and share both tangible and intangible resources, facilitating innovation and driving market expansion (Afeltra et al., 2023).

2.2.5. Collaboration and Co-creation of value

In a business ecosystem, collaboration and co-creation of value are driven by synergies and shared goals, allowing members to learn from one another, promote resource sharing, integration, and innovation, and ultimately benefit from co-creation activities (Babu et al., 2020). In such a complex, decentralized, and highly competitive environment, businesses must align their motivations and goals while accurately understanding the relationships and roles of their partners (Möller et al., 2020). New business ecosystems should leverage co-creation to unlock the potential of their members, a process that can be evaluated through factors such as resource richness, knowledge spillovers, direct and indirect externalities, and resource derivation (Zimmerling and Chen, 2021). Additionally, open information platforms provide stakeholders with more diversified and flexible ways to access and utilize resources (Thomas & Ritala, 2022).

Furthermore, collaboration and co-creation among members, combined with new technologies, can significantly enhance productivity and competitiveness (Senyo et al., 2019). By employing crowdsourcing to mitigate risks within the business ecosystem, resilience can be greatly strengthened. Therefore, a healthy business ecosystem should be built on sustainability, adaptability, innovation, and stability. Most importantly, innovation should be driven by the integration, exchange, and practical application of resources among ecosystem members (Park & Gil-Garcia, 2022).

3. The novel group decision-making approach: DIN-DEMATEL

This study combines the Delphi method and the interval single-valued trapezoidal neutrosophic DEMATEL technique as analytical tools. The primary objective is to leverage expert evaluations to assess the interrelationships among the dimensions and indicators. The interval single-valued trapezoidal neutrosophic numbers (SVTNN) capture and reflect the experts' hesitation and uncertainty in their judgments (Smarandache, 1999). To synthesize and integrate the diverse expert opinions, the Delphi method is employed to form a consensus and establish a judgment interval. This enhanced version clarifies the integration of the Delphi method and SVTNN, emphasizing the purpose and function of each tool in addressing expert hesitation and uncertainty. The explanation provides a clear structure for how the analysis was conducted.

Since the framework consists of two hierarchical levels—dimensions and indicators—and the dimensions are divided into five groups of indicators, executing the DIN-DEMATEL process requires six executions. This includes one execution at the dimension level and five executions at the indicator level, with each group of indicators being assessed separately.

The detailed research process is illustrated in Fig. 1, and the key steps involved are explained as follows.

Step 1. Define the linguistic variables for DIN-DEMATEL.

In the SVTNN, three key parameters are used: truth ($T_{\tilde{\theta}}$), falsity ($F_{\tilde{\theta}}$), and indeterminacy ($I_{\tilde{\theta}}$), with each parameter taking values within the range of 0–1. The flexibility of these three parameters, which are not strictly constrained, allows the model to adapt to varying degrees of uncertainty and ambiguity. This makes SVTNN particularly suitable for environments with high levels of fuzziness (Liang et al., 2018; Vafadarnikjoo et al., 2020).

A complete SVTNN consists of seven values, referred to as a neutrosophic set $\widetilde{\theta} = \{(\theta_1, \theta_2, \theta_3, \theta_4), (T_{\tilde{\theta}}, F_{\tilde{\theta}}, I_{\tilde{\theta}})\}$. This neutrosophic set is defined within the realm of real numbers, where truth-membership ($\mu_{\tilde{\theta}}(x)$), falsity-membership ($\lambda_{\tilde{\theta}}(x)$), and indeterminacy-membership ($\nu_{\tilde{\theta}}(x)$) correspond to the parameters of truth ($T_{\tilde{\theta}}$), falsity ($F_{\tilde{\theta}}$), and indeterminacy ($I_{\tilde{\theta}}$), respectively. The relationships and definitions of these parameters are detailed in Equations (1)–(3) (Chang et al., 2024; Lin & Lo, 2024).

$$\mu_{\tilde{\theta}}(x) = \begin{cases} \dfrac{(x-\theta_1)T_{\tilde{\theta}}}{(\theta_2-\theta_1)} & \theta_1 \le x \le \theta_2, \\ T_{\tilde{\theta}} & \theta_2 \le x \le \theta_3, \\ \dfrac{(\theta_4-x)T_{\tilde{\theta}}}{(\theta_4-\theta_3)} & \theta_3 \le x \le \theta_4, \\ 0 & otherwise. \end{cases} \tag{1}$$

$$\lambda_{\tilde{\theta}}(x) = \begin{cases} \dfrac{(\theta_2-x+F_{\tilde{\theta}}(x-\theta_1))}{(\theta_2-\theta_1)} & \theta_1 \le x \le \theta_2, \\ F_{\tilde{\theta}} & \theta_2 \le x \le \theta_3, \\ \dfrac{(x-\theta_3+F_{\tilde{\theta}}(\theta_4-x))}{(\theta_4-\theta_3)} & \theta_3 \le x \le \theta_4, \\ 1 & otherwise. \end{cases} \tag{2}$$

$$\nu_{\tilde{\theta}}(x) = \begin{cases} \dfrac{(\theta_2-x+I_{\tilde{\theta}}(x-\theta_1))}{(\theta_2-\theta_1)} & \theta_1 \le x \le \theta_2, \\ I_{\tilde{\theta}} & \theta_2 \le x \le \theta_3, \\ \dfrac{(x-\theta_3+I_{\tilde{\theta}}(\theta_4-x))}{(\theta_4-\theta_3)} & \theta_3 \le x \le \theta_4, \\ 1 & otherwise. \end{cases} \tag{3}$$

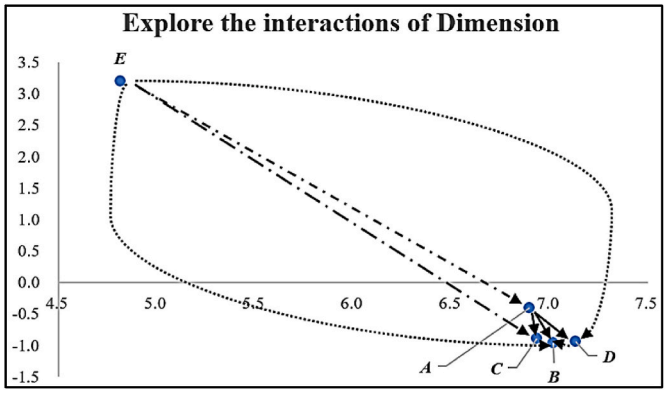


Fig. 1. The INRM of the dimensions.

The SVTNN structure can be divided into two main components. The first part is composed of trapezoidal fuzzy numbers $(\theta_1,\theta_2,\theta_3,\theta_4)$, where θ_1 represents the minimum value, θ_4 the maximum value, and it also includes the intermediate values θ_2 and θ_3, which do not necessarily need to be equal. The second part involves truth $(T_{\tilde{\theta}})$, falsity $(F_{\tilde{\theta}})$, and indeterminacy $(I_{\tilde{\theta}})$. Additionally, SVTNN can be defuzzified to obtain a single, precise value through Equation (4).

$$sc(\theta)=\frac{(\theta_1+2\theta_2+2\theta_3+\theta_4)(2+T_{\tilde{\theta}}-F_{\tilde{\theta}}-I_{\tilde{\theta}})}{18}. \tag{4}$$

When executing the DIN-DEMATEL process, experts use linguistic variables to quantify the interrelationships between indicators i and j, as shown in Table 2. The primary objective of this influence evaluation is to determine the degree of interrelationship among indicators. Experts also assess the reliability of these relationships based on the level of uncertainty, utilizing a 5-point Likert scale to generate the linguistic responses that will guide the results.

Step 2. Construct the SVTNNs direct relationship matrix.

According to the conversion rules in Table 2, the linguistic direct relationship matrix can be converted into an SVTNNs direct relationship matrix, as shown in Equation (5). $\otimes d_{ij}^{(s)}$ represents the evaluation rating given by expert s for the impact of indicator i on indicator j.

$$\otimes \boldsymbol{D}=\left[\otimes d_{ij}^{(s)}\right]_{n\times n}=\begin{bmatrix} \otimes d_{11}^{(s)} & \otimes d_{12}^{(s)} & \cdots & \otimes d_{1j}^{(s)} & \cdots & \otimes d_{1n}^{(s)} \\ \otimes d_{21}^{(s)} & \otimes d_{22}^{(s)} & \cdots & \otimes d_{2j}^{(s)} & \cdots & \otimes d_{2n}^{(s)} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \otimes d_{i1}^{(s)} & \otimes d_{i2}^{(s)} & \cdots & \otimes d_{ij}^{(s)} & \cdots & \otimes d_{in}^{(s)} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \otimes d_{n1}^{(s)} & \otimes d_{n2}^{(s)} & \cdots & \otimes d_{nj}^{(s)} & \cdots & \otimes d_{nn}^{(s)} \end{bmatrix}_{n\times n}, \tag{5}$$

$$i=j=1,2,\ldots,n;\ s=1,2,\ldots,q.$$

where $\otimes d_{ij}^{(s)}=\left\{\left(d_{ij,\theta_1}^{(s)},d_{ij,\theta_2}^{(s)},d_{ij,\theta_3}^{(s)},d_{ij,\theta_4}^{(s)}\right),\left(d_{ij,T_{\tilde{\theta}}}^{(s)},d_{ij,F_{\tilde{\theta}}}^{(s)},d_{ij,I_{\tilde{\theta}}}^{(s)}\right)\right\}$.

Step 3. Defuzzified the SVTNNs to generate a general form of the direct relationship matrix.

SVTNNs can be defuzzified into a general form of the direct relationship matrix using Equation (6), as shown in Equation (6).

$$\boldsymbol{D}=\left[d_{ij}^{(s)}\right]_{n\times n},\ i=j=1,2,\ldots,n;\ s=1,2,\ldots,q. \tag{6}$$

Step 4. Apply the Delphi method to integrate expert judgments and obtain the final direct relationship matrix.

Table 2
The linguistic variable of DIN-DEMATEL (Lo, 2024).

Influence evaluation part		Neutrosophic part	
Linguistic variable	Fuzzy Numbers	Linguistic variable	Fuzzy Numbers
Absolutely high influence (AI)	(0.8, 0.9, 0.9, 1)	Absolutely high confidence (AC)	(1.0, 0.0, 0.0)
High influence (HI)	(0.6, 0.7, 0.7, 0.8)	High confidence (HC)	(0.9, 0.1, 0.1)
Medium influence (MI)	(0.4, 0.5, 0.5, 0.6)	Medium confidence (MC)	(0.8, 0.1, 0.0)
Low influence (LI)	(0.2, 0.3, 0.3, 0.4)	Low confidence (LC)	(0.6, 0.2, 0.2)
No influence (NI)	(0, 0.1, 0.1, 0.1)	No confidence (NC)	(0.0, 0.0, 0.0)

To achieve a high level of group consensus amidst diverse expert judgments, the Delphi method is employed as a robust mechanism to refine and align expert evaluations. Initially, the experts' evaluations are sorted from smallest to largest. To eliminate potential biases and outliers, the lowest 20 % and highest 20 % of the data are systematically trimmed. This trimming process ensures that extreme opinions, which could skew the results, are excluded from further analysis. The remaining evaluations are then used to construct the final direct relationship matrix ($\boldsymbol{A}=\left[a_{ij}\right]_{n\times n}$). By identifying the maximum and minimum values within the refined data set and averaging them, the process achieves a balanced and representative consensus. This iterative approach not only mitigates the influence of individual biases but also fosters a collective agreement that reflects the expertise of the group.

Step 5. Execute the DEMATEL calculation procedure to derive the total influence relationship matrix and determine the indicators weights.

According to the DEMATEL calculation procedure (refer to Lo (2024)), the total influence relationship matrix ($\boldsymbol{T}=\left[t_{ij}\right]_{n\times n}$) can be obtained. On the other hand, the weights of the indicators can be calculated using **Equations (7)–(10)**.

$$(r_i)_{n\times 1}=\left[\sum_{j=1}^{n}t_{ij}\right]_{n\times 1}=(r_1,r_2,\ldots,r_i,\ldots,r_n), \tag{7}$$

$$(s_i)_{n\times 1}=\left[\sum_{i=1}^{n}t_{ij}\right]_{1\times n}^{transpose}=(s_1,s_2,\ldots,s_i,\ldots,s_n), \tag{8}$$

$$w_i=\frac{r_i+s_i}{\sum_{i=1}^{n}(r_i+s_i)}. \tag{9}$$

4. Data demonstration

For this study, which analyzes the interrelationships of evaluation indicators in the open data services industry's efforts toward digital transformation, multiple experts were invited to participate in discussions. Ultimately, 10 representative experts were selected for in-depth interviews, resulting in 10 completed DIN-DEMATEL questionnaires. These experts are actively involved in the open data services industry and have previously served as speakers on topics related to digital transformation. The group comprises 10 senior executives, all of whom are high-level leaders currently working or formerly employed by publicly listed companies. Their insights primarily reflect the perspectives of large enterprises, making the findings particularly relevant for organizations with substantial operational scales. Additionally, these experts include representatives from organizations and government agencies engaged in open data research and the development of business ecosystems. The detailed backgrounds of these experts are provided in Table 3.

Each expert was required to complete the questionnaire six times (once for the dimension level and five times for the indicator level). Here, we provide an explanation based on the analysis of the dimension level. Table 4 presents the evaluation results from Expert 1 at the dimension level. Expert 1 evaluated the influence and confidence between Dimension A and Dimension B as [HI, AC], which indicates that Dimension A has "high influence" on Dimension B, and the expert has a "absolutely high confidence" in this evaluation. Similarly, the expert evaluated the influence and confidence between Dimension B and Dimension A as [LI, AC], meaning that Dimension B has "low influence" on Dimension A, with the expert having "absolutely high confidence" in this assessment. This evaluation demonstrates that the influence assessment is not necessarily symmetric—i.e., the influence of Dimension A on Dimension B is not necessarily equal to the influence of Dimension B on Dimension A. The influence and confidence levels

Table 3
The detailed backgrounds of the 10 experts.

Number	Position	Organization	Years of Experience
Expert 1	COO	Data Service Company	18 years
Expert 2	General Manager	Data Service Company	15 years
Expert 3	Manager	Industry Association	19 years
Expert 4	General Manager	Data Service Company	20 years
Expert 5	Special Assistant	Data Service Company	7 years
Expert 6	Division Head	Research Organization	29 years
Expert 7	General Manager	Data Service Company	13 years
Expert 8	Vice General Manager	Data Service Company	27 years
Expert 9	General Manager	Data Service Company	30 years
Expert 10	Founder	Data Service Company	8 years

Table 4
The linguistic direct relationship matrix of dimension part: Expert 1.

	A	B	C	D	E
A	0	[HI, AC]	[AI, AC]	[MI, HC]	[MI, MC]
B	[LI, AC]	0	[HI, MC]	[AI, HC]	[AI, MC]
C	[LI, AC]	[MI, MC]	0	[AI, AC]	[AI, HC]
D	[MI, AC]	[HI, HC]	[LI, MC]	0	[AI, AC]
E	[MI, AC]	[HI, NC]	[MI, NC]	[AI, AC]	0

between different indicators may vary depending on the expert's perspective and confidence. The same logic applies to other evaluations, which can be inferred accordingly.

Next, using the linguistic variables of DIN-DEMATEL (as shown in Table 2), the responses from experts in Table 4 are converted into SVTNNs, forming the SVTNNs direct relationship matrix, as shown in Table 5. For example, when Expert 1 responds to Dimension A's influence on Dimension B as [HI, AC], the SVTNN linguistic conversion table is used to transform this into the corresponding semantic variable [(0.6, 0.7, 0.7, 0.9), (1, 0, 0)]. Following this, the final defuzzified value (crisp value) is calculated using Equation (4). For instance, in the case where the expert's response is [HI, AC], using these semantic variables, the final value is computed to be 0.717. This process is applied to each response, converting the results accordingly.

As this process advances, 10 SVTNNs direct relationship matrices will be generated, corresponding to each expert. However, to enhance the reliability of these evaluations, the Delphi method is applied to eliminate the influence of extreme values (as shown in Table 6). In this process, the top 20 % and bottom 20 % of data from the expert evaluations are removed, effectively discarding the extreme data points to prevent them from disproportionately affecting the results. After removing the extreme values, the maximum and minimum values within

Table 5
The SVTNNs direct relationship matrix of dimension part: Expert 1.

	A	B	…	E
A	0	[(0.6, 0.7, 0.7, 0.9), (1, 0, 0)]	…	[(0.3, 0.5, 0.5, 0.7), (0.8, 0.1, 0)]
B	[(0.1, 0.3, 0.3, 0.5), (1, 0, 0)]	0	…	[(0.8, 0.9, 0.9, 1), (0.8, 0.1, 0)]
C	[(0.1, 0.3, 0.3, 0.5), (1, 0, 0)]	[(0.3, 0.5, 0.5, 0.7), (0.8, 0.1, 0)]	…	[(0.8, 0.9, 0.9, 1), (1, 0, 0)]
D	[(0.3, 0.5, 0.5, 0.7), (1, 0, 0)]	[(0.6, 0.7, 0.7, 0.9), (0.9, 0.1, 0.1)]	…	[(0.8, 0.9, 0.9, 1), (1, 0, 0)]
E	[(0.3, 0.5, 0.5, 0.7), (1, 0, 0)]	[(0.6, 0.7, 0.7, 0.9), (1, 0, 0)]	…	0

Table 6
The Delphi method procedure of dimension part.

	A	B	…	E
A	0	[0.72, ~~0.3~~, 0.72, 0.65, 0.72, ~~0.3~~, 0.65, ~~0.81~~, 0.45, ~~0.81~~]	…	[0.45, ~~0.92~~, ~~0.1~~, ~~0.1~~, ~~0.92~~, 0.92, 0.1, 0.1, 0.12, 0.12]
B	[~~0.3~~, ~~0.27~~, 0.5, 0.5, ~~0.72~~, 0.5, ~~0.65~~, 0.45, 0.5, 0.9]	0	…	[0.81, 0.92, ~~0.12~~, ~~0.94~~, ~~0.94~~, ~~0.12~~, 0.17, 0.17, 0.17, 0.92]
C	[0.3, ~~0.27~~, ~~0.27~~, 0.65, ~~0.9~~, 0.3, 0.72, 0.72.3, ~~0.9~~]	[~~0.45~~, 0.65, ~~0.27~~, 0.72, 0.72, 0.5, 0.72, ~~0.9~~, 0.5, ~~0.9~~]	…	[~~0.19~~, ~~0.18~~, 0.18, 0.17, 0.12, 0.12, ~~0.11~~, ~~0.11~~, 0.12, 0.112]
D	[0.5, ~~0.3~~, 0.5, 0.5, ~~0.72~~, ~~0.72~~, 0.5, 0.72, ~~0.3~~, 0.65]	[0.65, ~~0.72~~, ~~0.72~~, ~~0.5~~, 0.65, 0.65, 0.65, 0.65, 0.65, ~~0.27~~]	…	[0.98, ~~0.99~~, 0.99, 0.94, 0.94, 0.94, ~~0.99~~, 0.99, ~~0.1~~, 0.1]
E	[0.5, ~~0.27~~, ~~0.27~~, 0.45, 0.65, ~~0.81~~, ~~0.65~~, 0.65, 0.65, 0.65]	[0.72, 0.72, 0.5, ~~0.45~~, 0.81, 0.65, 0.81, ~~0.81~~, ~~0.9~~, ~~0.27~~]	…	0

the remaining data range are identified. These values are then used to form the ISVTNN total direct relationship matrix, as shown in Table 7.

After obtaining the final direct relationship matrix, a standard DEMATEL analysis is performed to derive the total influence matrix, as shown in Table 8. This matrix not only considers the direct and indirect influences among the indicators but also measures the degree of self-influence. Additionally, through calculations, the degree of influence exerted by each indicator (r_i) and the degree to which each indicator is influenced (s_i) can be determined.

The results of the DEMATEL weight calculation are presented in Table 9. By analyzing the total influence ($r_i + s_i$) and net influence (r_i - s_i) between the indicators, the overall impact relationships are identified. Finally, based on the total influence ($r_i + s_i$), the weight of each indicator is calculated, and the indicators are ranked accordingly, as shown in Equations (7)–(9). The top dimension is "Innovation and Market Expansion (D)" at 0.218. In terms of the individual indicators, the top five weights are "Resource Utilization Efficiency (B1)" at 0.0573, "Innovation-Driven Productivity (B3)" at 0.0529, "Financial Performance and Sustainability (B4)" at 0.0526, "Value Creation Productivity (B2)" at 0.0512, and "Visionary Leadership (D1)" at 0.0471. These top five indicators are critical within the framework, and companies should prioritize them when making improvements, undergoing transformation, and formulating strategies.

Table 10 compares several DEMATEL-based methods, including original DEMATEL, interval DEMATEL, fuzzy DEMATEL, and neutrosophic DEMATEL. The proposed DIN-DEMATEL method offers significant advantages over past DEMATEL approaches. Integrating the

Table 7
The final direct relationship matrix of dimension part.

*Interval value

	A	B	C	D	E
A	0	[0.450, 0.717]	[0.500, 0.810]	[0.450, 0.900]	[0.098, 0.100]
B	[0.450, 0.645]	0	[0.500, 0.717]	[0.450, 0.717]	[0.098, 0.117]
C	[0.300, 0.717]	[0.500, 0.717]	0	[0.500, 0.717]	[0.116, 0.117]
D	[0.500, 0.717]	[0.645, 0.645]	[0.450, 0.645]	0	[0.094, 0.100]
E	[0.450, 0.645]	[0.500, 0.810]	[0.500, 0.717]	[0.450, 0.900]	0

Crisp value

	A	B	C	D	E
A	0.000	0.584	0.655	0.675	0.099
B	0.548	0.000	0.609	0.584	0.108
C	0.509	0.609	0.000	0.609	0.117
D	0.609	0.645	0.548	0.000	0.097
E	0.548	0.655	0.609	0.675	0.000

Table 8
The total influence matrix.

*Dimension part					
	A	B	C	D	E
A	0.581	0.819	0.827	0.850	0.167
B	0.720	0.591	0.774	0.784	0.161
C	0.709	0.783	0.579	0.789	0.163
D	0.748	0.806	0.771	0.611	0.160
E	0.894	0.988	0.960	1.002	0.159
** Indicator part of dimension A*					
	A1	A2	A3	A4	A5
A1	1.830	1.648	1.928	2.058	1.926
A2	1.950	1.399	1.829	1.916	1.838
A3	2.022	1.628	1.717	2.026	1.945
A4	1.653	1.331	1.540	1.468	1.534
A5	1.734	1.372	1.653	1.716	1.470
** Indicator part of dimension B*					
	B1	B2	B3	B4	
B1	2.463	2.254	2.443	2.787	
B2	2.638	1.961	2.362	2.638	
B3	2.597	2.139	2.097	2.632	
B4	2.253	1.848	2.001	2.056	
** Indicator part of dimension C*					
	C1	C2	C3	C4	C5
C1	1.903	1.709	2.007	2.114	2.037
C2	2.092	1.512	1.997	2.048	2.000
C3	2.031	1.660	1.756	2.025	1.986
C4	1.713	1.371	1.611	1.519	1.614
C5	1.831	1.436	1.747	1.814	1.577
** Indicator part of dimension D*					
	D1	D2	D3	D4	D5
D1	1.787	1.511	1.847	2.023	1.929
D2	1.849	1.248	1.727	1.844	1.794
D3	1.928	1.450	1.595	1.920	1.876
D4	1.640	1.236	1.502	1.458	1.557
D5	1.777	1.299	1.642	1.760	1.518
** Indicator part of dimension E*					
	E1	E2	E3	E4	E5
E1	1.588	1.373	1.745	1.827	1.764
E2	1.691	1.139	1.655	1.683	1.661
E3	1.707	1.302	1.469	1.719	1.695
E4	1.465	1.115	1.400	1.290	1.415
E5	1.529	1.134	1.492	1.513	1.317

Delphi method to eliminate extreme opinions and utilizing interval neutrosophic fuzzy values effectively addresses uncertainty and ambiguity in expert evaluations. Unlike earlier methods, DIN-DEMATEL considers uncertainty and response confidence, ensuring more balanced and reliable results. Tailored for the significant data era, this approach provides robust, actionable insights, making it highly applicable for complex decision-making and enhancing the practicality and precision of industry strategies.

5. Discussion and conclusion

This section discusses the managerial implications of the study, providing actionable insights for practitioners in the open data services industry. Additionally, this section concludes the research by summarizing its main contributions, addressing study limitations, and proposing future research directions.

5.1. Management implications

The analysis presented in Section 4 shows several key managerial implications tailored to large enterprises. These implications provide valuable insights aimed explicitly at addressing large-scale organizations' unique challenges and opportunities in the open data services industry. They offer practical guidance on enhancing operational efficiency, fostering innovation, and improving strategic decision-making, ensuring that the findings are particularly relevant for organizations with substantial resources and complex operational structures.

Table 9
The results of the DIN-DEMATEL.

Dimension	Local Weight	Ranking	Indicator	Local Weight	Global Weight	Ranking
A	0.210	4	A1	0.215	0.0453	7
			A2	0.189	0.0398	18
			A3	0.209	0.0439	9
			A4	0.194	0.0407	14
			A5	0.193	0.0406	17
B	0.214	2	B1	0.268	0.0573	1
			B2	0.239	0.0512	4
			B3	0.247	0.0529	2
			B4	0.246	0.0526	3
C	0.211	3	C1	0.214	0.0453	6
			C2	0.192	0.0406	16
			C3	0.206	0.0435	10
			C4	0.192	0.0407	15
			C5	0.195	0.0413	13
D	0.218	1	D1	0.217	0.0471	5
			D2	0.182	0.0396	19
			D3	0.205	0.0445	8
			D4	0.197	0.0427	12
			D5	0.200	0.0435	11
E	0.147	5	E1	0.216	0.0317	20
			E2	0.184	0.0271	24
			E3	0.208	0.0305	21
			E4	0.195	0.0287	23
			E5	0.197	0.0289	22

According to the analysis (Fig. 1), Data Governance and Transparency (A) and Collaboration and Co-Creation of Value (E) are key drivers influencing Operational Efficiency (B), System Resilience (C), and Innovation and Market Expansion (D). Effective data governance ensures high data quality, security, and risk mitigation, directly enhancing operational efficiency and resilience. Meanwhile, strong collaboration fosters partnerships and resource-sharing, driving innovation and enabling market expansion. The findings suggest that robust data governance provides a foundation for informed decision-making and risk management, ensuring optimized processes and minimized risks. Organizations prioritizing collaboration can leverage external resources, knowledge, and capabilities, accelerating innovation and market growth. The synergy between data management and collaboration highlights their critical role in enhancing efficiency, resilience, and innovation. Leaders should strengthen data governance practices and build collaborative ecosystems to maximize strategic outcomes. Businesses can improve operational effectiveness, mitigate risks, and drive sustainable innovation in an increasingly data-driven and interconnected market.

Within Data Governance and Transparency (A) (Fig. 2), Data Quality (A1), Government Policies (A2), and Data Interoperability (A3) are key factors influencing Data Circulation (A4) and Demand Alignment (A5). High-quality data ensures accuracy and reliability, enabling seamless data flow. This aligns with the findings of Janssen et al. (2020), who emphasize that data quality, including accuracy, timeliness, and completeness, is crucial for trustworthy Big Data Algorithmic Systems (BDAS). Government policies establish regulatory frameworks for data sharing and access, impacting circulation and alignment with user demands. As highlighted in multiple studies, government regulations and public sector-specific requirements significantly affect technology adoption and data governance (Gegenhuber et al. (2023)). Data interoperability facilitates integration across systems, ensuring data effectively meets stakeholder needs. The importance of data interoperability is further supported by the necessity for trusted data-sharing frameworks that guide data exchange within and between participating organizations (Janssen et al., 2020). The findings highlight that enhancing

Table 10
Comparison of various DEMATEL-Based methods.

	Elimination of Extreme Opinions	Integration of Expert Opinions	Data Type	Consideration of Uncertainty	Consideration of Confidence in Responses
Original DEMATEL	No	Arithmetic Mean	Crisp Value	No	No
Interval DEMATEL	No	Maximum and Minimum Values	Interval Value	Yes	No
Fuzzy DEMATEL	No	Arithmetic Mean	Fuzzy Value	Yes	No
Neutrosophic DEMATEL	No	Arithmetic Mean	Neutrosophic Fuzzy Value	Yes	Yes
DIN-DEMATEL (Our Proposed)	Yes, using Delphi method to eliminate top and bottom 20 % extreme opinions	Maximum and Minimum Values	Interval Neutrosophic Fuzzy Value	Yes	Yes

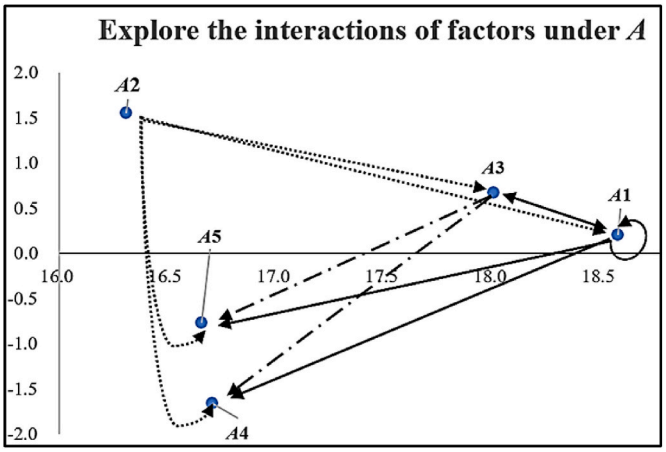


Fig. 2. The INRM of the indicators under "Data Governance and Transparency (A)" dimension.

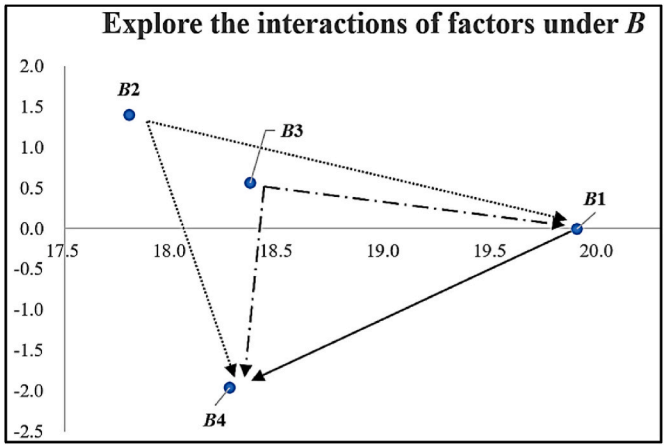


Fig. 3. The INRM of the indicators under "Operational Efficiency and Productivity Enhancement (B)" dimension.

data quality, regulatory compliance, and interoperability is crucial for optimizing data circulation and meeting market demands. Organizations that ensure reliable and seamlessly exchangeable data foster better information flow, enhancing its relevance and usability. Managers should prioritize these governance elements to improve data accessibility and strategic alignment. Moreover, establishing clear data governance spaces with well-defined decision-making authority, rights, roles, and responsibilities is essential for managing data effectively across multiple actors (Paparova et al., 2023). Strengthening data quality controls, adapting to policy changes, and enabling system interoperability will enhance market responsiveness, operational efficiency, and informed decision-making, ultimately driving business competitiveness in a data-driven economy. Additionally, fostering a culture that treats data as an asset and supports data-driven decisions is critical, as is addressing the challenges of transparency and accountability in open data policies (Park & Gil-Garcia, 2022).

Within Operational Efficiency and Productivity Enhancement (B) (Fig. 3), Resource Utilization Efficiency (B1), Value Creation Productivity (B2), and Innovation-Driven Productivity (B3) significantly impact Financial Performance and Sustainability (B4). Efficient resource management maximizes output while minimizing costs, directly improving financial stability. This aligns with findings highlighting the importance of efficient resource use and cost reduction for organizational performance (Schneider et al., 2023). Value creation productivity enhances profitability by increasing market competitiveness, while innovation-driven productivity fosters new solutions, boosting operational efficiency and revenue generation. As Kitsios and Kamariotou (2023) noted, innovative business models can be a source of competitive advantage, transforming processes, products, or services to increase value. The findings suggest optimizing resource use, fostering innovation, and focusing on value creation are key to strengthening financial performance and long-term sustainability. Businesses that efficiently manage resources can lower operational costs and improve financial health.

Meanwhile, developing innovative products and services differentiates companies in the marketplace, driving growth and profitability. Innovation-driven productivity enhances adaptability, allowing organizations to respond to market shifts and seize new opportunities. Managers should prioritize continuous innovation, resource optimization, and value-driven strategies to ensure sustained financial success and competitiveness in an evolving business environment. Specifically, organizations should focus on strategic management practices to allocate resources effectively and address resistance to change, which is critical for successful AI adoption and improved organizational performance (Matheus et al., 2020).

Within System Resilience and Risk Management (C) (Fig. 4),

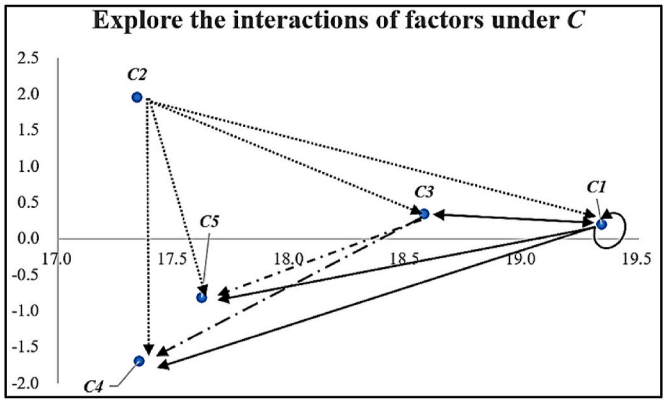


Fig. 4. The INRM of the indicators under "System Resilience and Risk Management (C)" dimension.

Predictive Capability (C1), Network Stability and Structural Robustness (C2), and Adaptive Capacity (C3) play a crucial role in enhancing Recovery Capability (C4) and Strategic Flexibility and Timely Adjustment (C5). Predictive capability allows organizations to anticipate risks, while network stability ensures resilience under pressure. Adaptive capacity enables quick responses to unexpected challenges, collectively strengthening recovery and strategic flexibility. These factors are interrelated—improvements in one reinforce the others—highlighting the need for a holistic resilience strategy. Predictive capability helps foresee disruptions, but its effectiveness depends on a stable network (C2) that can absorb shocks. Additionally, adaptive capacity (C3) ensures organizations can pivot quickly, reinforcing recovery (C4) and strategic adjustment (C5). The findings emphasize the importance of an integrated risk management approach where predictive analytics, network stability, and adaptability function together. Managers should develop these capabilities simultaneously to enhance resilience, enabling organizations not only to recover from disruptions but also to proactively adjust strategies in dynamic environments.

Recent literature has provided substantial support. For example, Robertson et al. (2020) emphasized that public sector support is crucial for establishing entrepreneurial ecosystems, which aligns with the perspective that predictive capabilities enable organizations to anticipate risks. Public sector support can be viewed as an external resource that helps enterprises predict market changes and risks, enhancing overall resilience. Neumann et al. (2024) indicated that organizational culture and external pressures (such as government regulations) play vital roles in driving AI adoption, corresponding with the emphasis on network stability and adaptive capacity. Stable organizational networks and cultures are better positioned to adapt to external pressures and swiftly adopt new technologies (such as AI) to enhance competitiveness.

Within Innovation and Market Expansion (D) (Fig. 5), Visionary Leadership (D1), Proactive Engagement (D2), and Product and Service Differentiation (D3) significantly impact Leverage of Intangible Assets (D4) and Integration with External Networks (D5). Visionary leadership sets the strategic direction for innovation, enabling organizations to maximize intangible assets like intellectual property and brand equity. Proactive engagement fosters collaboration, strengthening integration with external networks, while differentiated products and services enhance market positioning, making partnerships more valuable. The findings highlight that strong leadership, stakeholder engagement, and product innovation are key to leveraging intangible assets and building external collaborations. Visionary leadership identifies opportunities to utilize brand reputation and intellectual capital, while proactive engagement ensures active stakeholder participation, facilitating smoother integration with external networks. Additionally, product differentiation attracts strategic partners, expanding opportunities for resource exchange and market influence. To fully harness intangible assets and establish strong partnerships, managers should prioritize innovative leadership, stakeholder involvement, and continuous differentiation of offerings. These efforts enhance competitive advantage, drive innovation, and create sustainable market expansion strategies.

Within Collaboration and Co-Creation of Value (E) (Fig. 6), Responsiveness to Market Demands (E1), Resource Complementarity (E2), and Knowledge and Experience Sharing (E3) significantly impact Resource Integration Capability (E4) and Collaborative Value Proposition (E5). Responsiveness to market demands enables companies to swiftly adapt to customer needs, ensuring effective resource integration. Resource complementarity strengthens partnerships by aligning diverse capabilities, maximizing value creation. Knowledge and experience sharing deepens collaboration, fostering a stronger value proposition. The findings emphasize that agility, resource alignment, and knowledge exchange are key to enhancing collaboration and co-creation. Responsiveness allows businesses to integrate resources dynamically, meeting evolving customer expectations. Resource complementarity ensures that partners leverage unique strengths for more efficient integration. Additionally, shared knowledge strengthens relationships, leading to innovative and competitive value propositions. To enhance collaboration and maximize value creation, managers should cultivate an ecosystem that prioritizes resource sharing, agility, and market-driven innovation. A proactive approach to integrating complementary assets and acting on market feedback will not only improve operational efficiency but also drive sustainable competitive advantages in partnerships and co-creation efforts.

Previous literature has provided support for the arguments mentioned above. Kitsios and Kamariotou (2023) indicated that open data hackathons facilitate collaboration among multiple stakeholders (including businesses, governments, and research institutions). Such collaboration enables participants to share knowledge, jointly develop innovative applications, and create new business models, aligning with the perspective of knowledge and experience sharing (E3). Neumann et al. (2024) emphasized that organizational collaboration with external partners can help acquire resources and skills that organizations lack internally, particularly in AI applications. Partnerships with professional service providers or academic institutions can promote knowledge transfer and resource integration, echoing the concept of resource complementarity (E2).

Slimani et al. (2024) noted that the proliferation of ICT (Information and Communication Technology) facilitates economic actors' access to key markets and the development of their activities, corresponding to market demand response (E1). Digital platforms also simplify economic actors' access to essential markets and business expansion, thereby increasing foreign direct investment and remittances. Janssen et al.

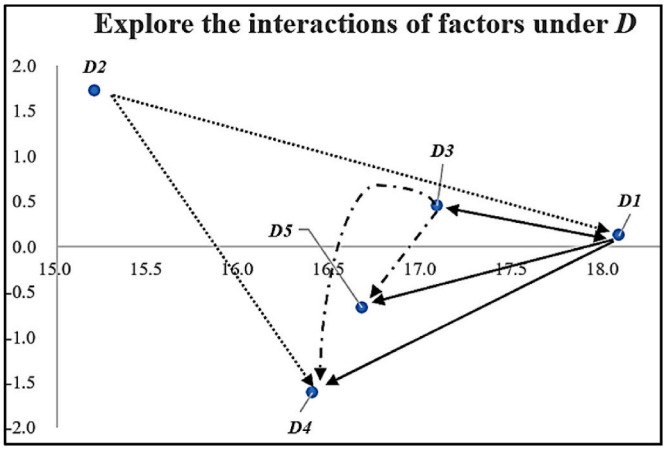


Fig. 5. The INRM of the indicators under "Innovation and Market Expansion (D)" dimension.

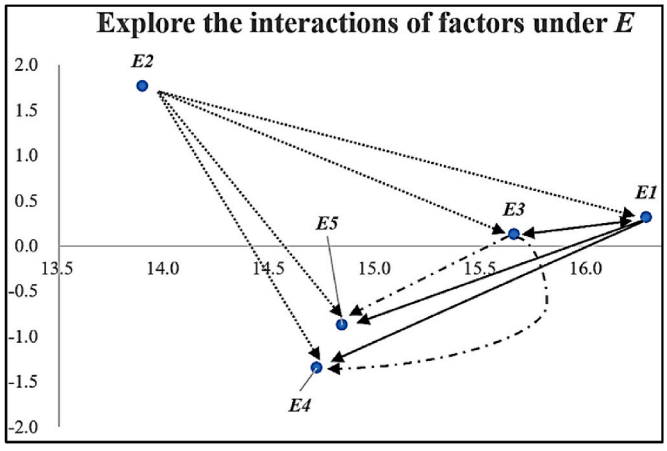


Fig. 6. The INRM of the indicators under "Collaboration and Co-Creation of Value (E)" dimension.

(2020) emphasized that data governance, transparency, collaboration, and value co-creation are key drivers affecting operational efficiency, system resilience, innovation, and market expansion. This resonates with the view of cultivating ecosystems through resource sharing, flexibility, and market-driven innovation to enhance collaboration and maximize value creation. These research findings collectively reinforce the importance of collaboration and value co-creation while providing empirical support.

In the overall evaluation system, the three most influential indicators are Resource Utilization Efficiency (B1), Innovation-Driven Productivity (B3), and Financial Performance and Sustainability (B4), all of which are critical to the operational and strategic success of large organizations. These indicators not only hold theoretical significance but also have strong practical applications.

Resource Utilization Efficiency (B1) maximizes output while minimizing costs by optimizing labor, capital, and technology. For example, a multinational manufacturing company improved efficiency by adopting advanced analytics and automation, reducing waste by 15 % and achieving substantial cost savings. Innovation-Driven Productivity (B3) emphasizes continuous development of products, services, and processes to maintain competitiveness. A global technology firm integrated AI-driven R&D solutions, significantly reducing time to market while fostering cross-functional collaboration and enhancing its market position. Financial Performance and Sustainability (B4) ensures long-term viability. A leading retail chain improved financial performance by implementing strategic cost control, optimizing cash flow, and investing in sustainability initiatives such as energy-efficient operations, strengthening its brand reputation and customer loyalty.

The findings of this study align with and extend previous research, providing both consistency and novel insights. The emphasis on Resource Utilization Efficiency (B1) and Innovation-Driven Productivity (B3) as key factors for operational success is consistent with existing research that highlights the role of efficient resource management and continuous innovation in maintaining competitiveness (Wen et al., 2023). Previous studies have also underscored the importance of optimizing resource allocation and leveraging emerging technologies, which our study corroborates by identifying specific implementation strategies (Neumann et al., 2024). However, our findings go further by revealing new dimensions, such as the integration of Financial Performance and Sustainability (B4). This aspect reflects a growing focus on long-term viability and sustainable development, contrasting with traditional studies that primarily emphasize short-term financial gains. By demonstrating the dual benefits of sustainable practices—such as cost reduction and enhanced brand reputation—our study offers a more comprehensive perspective.

Moreover, while previous studies have often examined these factors in isolation, our use of the DIN-DEMATEL methodology provides a novel contribution by uncovering the interrelationships and cascading effects among these indicators. For instance, the interplay between operational efficiency and innovation can be observed in companies like Amazon and Alibaba, which have leveraged advanced data analytics and automation to optimize warehouse management and last-mile delivery. These innovations not only improved resource utilization but also enhanced customer satisfaction and reduced operational costs, demonstrating how advancements in one area can amplify benefits in another. Similarly, Tesla's approach of integrating sustainable practices with financial planning—such as investing in energy-efficient manufacturing and optimizing supply chains—has not only boosted financial performance but also strengthened its brand reputation. These examples illustrate how a strategic focus on both efficiency and innovation can create a reinforcing cycle of improvements. This integrated approach not only deepens the understanding of key success factors but also offers actionable insights for optimizing performance in the open data services industry.

5.2. Summary and future research

This study explores the open data services industry's role in digital transformation and AI development. It addresses challenges in data governance, collaboration, and resilience and proposes a comprehensive evaluation framework to enhance industry competitiveness and support AI-driven innovation. While the findings are directly applicable to the open data services industry, the proposed evaluation framework and key insights could also be adapted to other industries undergoing digital transformation. For example, sectors such as healthcare, finance, and manufacturing, which face similar challenges related to data management, collaboration, and AI integration, could benefit from applying the identified indicators and relationships to optimize their operations and decision-making processes. However, it is important to acknowledge certain limitations in this regard. The industry-specific characteristics and regulatory environments of other sectors may influence the applicability and effectiveness of the proposed framework.

This study's significant contributions are threefold. First, it constructs a robust evaluation model that integrates multiple dimensions, providing a systematic understanding of critical factors. Second, by applying the Delphi method, the study successfully eliminates extreme opinions in group assessments, ensuring balanced and reliable expert consensus. This approach enhances the robustness of the findings by reducing subjectivity and bias. Third, the DIN-DEMATEL analysis identifies and clarifies key indicators' interrelationships and relative importance, offering actionable insights into how these factors impact business performance and drive AI-driven innovation. The study highlights the foundational role of data governance and collaboration in enabling businesses to adopt AI technologies effectively. Improved data management practices, as proposed, empower organizations to support AI systems better, enabling more accurate, transparent, and efficient AI applications. Finally, the research provides actionable recommendations for the open data services industry, guiding businesses as they enhance their competitive edge, operational practices, and adaptability in the digital transformation era. The insights and impact of this research are significant. This study offers a comprehensive framework for improving operational practices and fostering innovation by addressing the challenges of data governance, collaboration, and resilience. The findings provide a practical guide for businesses navigating digital transformation and contribute to the broader discourse on AI readiness and open data optimization, ensuring that companies remain at the forefront of technological advancements.

This study also acknowledges several limitations that may affect the generalizability of its findings. First, the data used in this research were primarily derived from a specific group of experts and literature reviews, which may limit the comprehensiveness of the identified indicators and their interrelationships. The insights gathered may not fully capture the diverse needs of different industries or regions. Second, while the evaluation framework utilizes the DIN-DEMATEL approach for group decision-making, it still relies heavily on expert opinions, which may introduce subjectivity into the findings. Additionally, the availability and quality of data in specific contexts may pose challenges to the practical application of the proposed model.

Future research could extend in several directions. First, further studies could validate the proposed model across different regions and industries by conducting cross-national or cross-sectoral comparisons to examine how the identified indicators perform under various conditions. Second, future research could incorporate more empirical data or big data analysis to enhance the objectivity and applicability of the evaluation framework, thereby reducing reliance on expert judgment. Additionally, as AI technologies evolve, future studies could explore the deeper integration of open data with AI applications, examining how open data can be more effectively leveraged to enhance AI performance. This includes developing forward-looking data governance strategies to address emerging data security and privacy challenges. Moreover, ethical considerations such as data privacy, security, and the potential

risks of AI misuse are critical issues that require further exploration within the open data services industry. Addressing these challenges comprehensively in future research could provide valuable insights into balancing the benefits of AI-driven innovation with the need for responsible data management practices. A more in-depth examination of cybersecurity measures, privacy-preserving techniques, and regulatory compliance could help organizations mitigate risks associated with AI applications and open data utilization.

Furthermore, future research could extend the DIN-DEMATEL approach by incorporating the HISA (Heterogeneous Influence and Strength Attenuation) concept, allowing for the assessment of both positive and negative influences rather than merely determining whether an impact exists. This enhancement would offer a more nuanced understanding of indicator relationships and improve decision-making accuracy. These efforts would not only validate the framework but also expand its relevance and applicability in a rapidly evolving digital landscape, ensuring that the open data services industry remains resilient and ethically responsible.

CRediT authorship contribution statement

Ya-Ting Chang: Writing – original draft, Validation, Formal analysis, Data curation, Conceptualization. **Huai-Wei Lo:** Writing – review & editing, Visualization, Resources, Methodology. **Sheng-Wei Lin:** Writing – review & editing, Writing – original draft, Validation, Funding acquisition, Formal analysis, Data curation.

Data availability statement

The research data involves confidential information from the companies, and as a result, not all survey data can be made publicly available.

Declaration of competing interest

All authors declare that they have no conflicts of interest. No human or animal experiments were conducted in this study.

Data availability

Data will be made available on request.

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